Highlights

A decision procedure for machine-proposed engineering numbers: three-valued verdicts over a declared uncertainty envelope, with witnesses and the threshold that decides[⋆]

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- A proposed number is decided over the whole declared envelope, not at a sample of it.
- Three verdicts with witnesses: proved, refuted, or refused with the threshold that decides.
- Monte Carlo found 0.3% exceedances and a worst draw of 360.7 bar; the proof reaches 361.6.
- Seven injected faults, each refuted or refused with its reason; none passes.
- The record re-runs without the authors' code; a second implementation agrees at 176 inputs.

A decision procedure for machine-proposed engineering numbers: three-valued verdicts over a declared uncertainty envelope, with witnesses and the threshold that decides

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Abstract

Digital twins, surrogate models and optimisers now propose operating and design numbers — an injection rate, a wellhead pressure, a design wave — and each arrives with a confidence that speaks of the model, not of that output. Verification and validation qualify the model; uncertainty quantification gives a dispersion; neither decides whether this number holds for the conditions the engineer has declared. This paper describes a decision procedure that does. The engineer declares a physical model, an envelope of inputs as intervals with their sources, and the operating rules; a proposer — any model, opened or not — submits a number and what it asserts; the procedure evaluates the declared model over the whole envelope in outward-rounded interval arithmetic and returns one of three verdicts for each check: PROVED (holds for every input in the envelope), REFUTED (fails, with a proof), or REFUSED (the envelope holds inputs on both sides of the rule, one proved on each side), together with the witnesses and the threshold on a declared quantity that would turn a refusal into a proof. The procedure is demonstrated on a water-injection flowline with the Colebrook and Darcy–Weisbach equations, where a plausible recommendation to raise injection by 4.2% is REFUSED: inputs in the declared envelope give wellhead pressures of 361.6 and 344.2 bar against a limit of 360 bar, and the recommendation becomes PROVED at a pump discharge of 204.0 bar or less. Seven injected faults of the kinds an industrial verifier must catch are each refuted or refused with the reason. A Monte Carlo study of the same recommendation

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finds exceedances in 0.3% of 10,000 uniform draws with a worst draw of 360.7 bar; the proved worst case is 361.6 bar, and the verdict does not depend on a prior over the envelope. The same grammar is then applied to numbers that were not produced for a demonstrator: maximum-likelihood extreme-value fits of significant wave height over a public hindcast, a scientific library's two answers for one 100-year wave, and a published design table. A second implementation in another language and arithmetic, sharing no code, agrees with the gate at 176 inputs, and an outside group has re-certified one of the project's records without its code. The contribution is the grammar and the re-runnable record, not the interval arithmetic that carries them.

Keywords: decision procedure, interval arithmetic, verification and validation, digital twin, uncertainty envelope, counterexample, management of change

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1. Introduction

A model that proposes a number has become ordinary in production engineering. Digital twins recommend set points, surrogate models replace simulators inside optimisation loops, and large language models now draft operating recommendations with a stated confidence. Operators report the gains in aggregate (Petrobras, 2026a,b); the obligation that follows each number is older: under process-safety regimes a change to an operating specification is a managed change, with a technical record and an integrity review (ANP, 2007). What the record should contain for a number a model proposed is the question this paper takes up.

Current practice answers it with two instruments, both aimed at the model. Verification and validation qualify a code and a model against reference solutions and experiments, and estimate numerical and validation uncertainty (ASME, 2009; NASA, 2016; Oberkampf and Roy, 2010; Roy and Oberkampf, 2011); the result is a statement about the model's credibility in a domain. Uncertainty quantification propagates input distributions to an output dispersion, by sampling or by surrogates, and reports a probability of exceedance; imprecise-probability methods widen the input description to intervals and probability boxes and propagate those (Ferson and Ginzburg, 1996; Beer et al., 2013). Neither instrument decides the question an engineer faces at the moment of acting: *for the inputs we have declared, does this number hold?* A credible model can be run outside its envelope; a sampled probability of exceedance of 0.3% says nothing about the input the sampler did not draw; and the proposer's own confidence is a property of the

proposer.

This paper describes a decision procedure that answers that question directly and records the answer so that anyone can re-run it. The ingredients are standard: a declared model, a declared box of inputs, interval arithmetic with outward rounding (Moore et al., 2009), and branch-and-bound or monotonicity arguments to bound an output over a box (Hansen and Walster, 2004; Jaulin et al., 2001; Rump, 2010). What is new is the grammar of the result and the discipline around it. Every check returns one of three verdicts — PROVED, REFUTED, REFUSED — and a refusal is not a failure of the method but a finding: the declared evidence does not decide, and here are an input on each side of the rule, each with a proved value, and the threshold on a declared quantity that would decide. The record of a decision carries the model, the envelope, the verdicts, the witnesses and the thresholds, and it is re-run by a second implementation that shares no code with the first. Refusal as a verdict is what turns the procedure from a bound into an instrument: the output of a refusal is a measurement to make or a parameter to disclose, priced by what it would decide.

Section 2 defines the grammar. Section 3 describes the instrument on a water-injection flowline, chosen as the pilot because single-phase, steady, isothermal flow is covered whole by the declared model, so that the limits of the declared model are a design choice rather than an omission. Section 4 demonstrates it on a proposed recommendation and seven injected faults, and sets a Monte Carlo study of the same recommendation beside it. Section 5 applies the same grammar to numbers that were not produced for a demonstrator: extreme-value fits of significant wave height over a public hindcast, a library's two answers for one design wave, and a published design table, which are the subject of a companion paper (Toledo, 2026). Section 6 places the procedure beside verification and validation, uncertainty quantification and formal methods; Section 7 discusses what a verdict means and what it costs.

2. The decision grammar

2.1. Objects

A *declared model* M is a set of equations with a stated domain — here Colebrook's equation for the friction factor in turbulent pipe flow and the Darcy–Weisbach pressure drop with the hydrostatic gain of a descending line. A *declared envelope* B is a box: each input as an interval with the source of its bounds (a pipe survey, a fluid analysis, a bathymetric chart). The *rules* R are inequalities on quantities the model computes, each with

its source — an operating limit on wellhead pressure, an erosional velocity limit (API, 1991). A *proposal* is a number with what it asserts: an operating point (flow rate Q, pump discharge P_d) and the output the proposer claims for it (a wellhead pressure P_{wh}), optionally with further asserted quantities (a velocity, a split of flow at a manifold). The proposer may be anything: an optimiser, a simulator, a spreadsheet, a language model. It need not be opened; the procedure decides its output.

2.2. Checks

A proposal is decided by a list of *checks*, each a statement about the declared model over the declared envelope:

- *physical bounds* that hold in any flow regime, such as a friction loss that cannot be negative, so that the claimed pressure cannot exceed the discharge plus the hydrostatic gain;
- *domain* conditions under which the declared model applies, such as a Reynolds number above the turbulent threshold for Colebrook's correlation;
- *rules*, the declared operating limits;
- *consistency*, whether the claimed output lies inside the enclosure the declared model gives over the envelope, and whether further asserted quantities (a velocity, a mass balance) are what the inputs give.

2.3. Verdicts

Each check returns one of three words:

proved the statement holds for every input in the envelope. For a rule, the enclosure of the quantity over the whole box lies on the allowed side; for a consistency check, the claim lies inside the enclosure.

refuted the statement fails with a proof: for every input in the envelope (the enclosure lies wholly on the forbidden side), or at a named impossibility (a claimed pressure above the physical bound, a mass balance that does not close by an exact amount, a claimed velocity outside what the inputs allow).

refused the declared evidence does not decide: the envelope holds inputs on both sides of the rule, one on each side with its value proved by a thin-box evaluation, and the procedure publishes those two *witnesses*

together with the *thresholds* — the values of a declared quantity (here the pump discharge, or the pipe roughness) at which the verdict over the unchanged envelope becomes PROVED or REFUTED.

The verdict of the whole receipt is REFUTED if any check refutes, otherwise REFUSED if any check refuses, otherwise PROVED. The three words are deliberately not probabilities: a refusal says that the question is open under the declared evidence, and names what would close it.

2.4. Rules and flags

A consistency check compares the proposer's number with a declared reference model. When the proposer's physics is richer than the declared model — a thermal profile, a multiphase correlation — a number outside the declared enclosure is not wrong; it is inconsistent with a cruder reference. The procedure therefore distinguishes two kinds of check. Rule checks and physical bounds are evaluated with bounding models that hold in any regime (a friction loss cannot be negative whatever the correlation), so their verdicts stand on their own. Consistency checks are flags for the engineer to arbitrate: in the demonstrator of Section 4 a consistency failure is reported as REFUTED relative to the declared model, and in a deployment the asset's engineer decides, when the model and the envelope are declared, which checks are rules and which are flags. The pilot domain was chosen so that the question does not arise: single-phase, steady, isothermal water injection is covered whole by Colebrook and Darcy–Weisbach, and the declared model is the proposer's own physics or a bound on it.

3. The instrument

3.1. The declared model

The pilot is a water-injection flowline from a floating production unit to an injection well in about 2 100 m of water (Table 1), with the physics an engineer would write on one page:

$$A = \tfrac{\pi}{4} D^2, \qquad v = Q/A, \qquad \mathrm{Re} = \rho v D/\mu, \tag{1}$$

$$\frac{1}{\sqrt{f}} = -2\log_{10}\left(\frac{\varepsilon}{3.7D} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right) \qquad \text{(Colebrook, 1939), turbulent flow only,} \tag{2}$$

$$P_{wh} = P_d + \rho g \Delta z - f\frac{L}{D}\frac{\rho v^2}{2}, \tag{3}$$

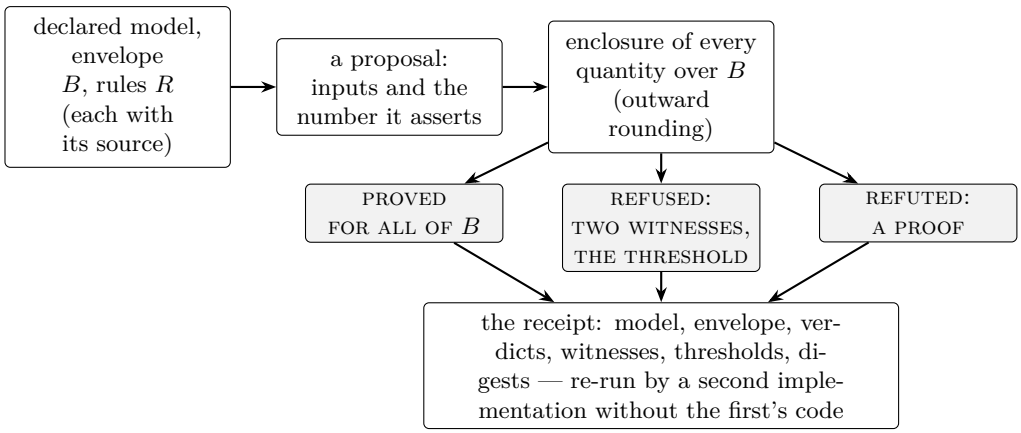


Figure 1: The decision grammar. The proposer is not opened; its number is decided over the declared envelope, and a refusal names what would decide it.

with g the standard gravity, exact by definition. A PROVED here means: for every input in the declared envelope, this model gives a wellhead pressure inside the enclosure and every declared rule holds. It is a statement about the declared model and the declared inputs, not about the sea.

3.2. The enclosure without a square root

Write $x = 1/\sqrt{f}$. Then x is the root of $h(x; a, b) = x + (2/\ln 10)\ln(a + bx)$ with $a = \varepsilon/(3.7D)$ and $b = 2.51/\mathrm{Re}$. For $x > 0$, $a \geq 0$ and $b > 0$, h is strictly increasing in x, in a and in b, so the root $x^*(a, b)$ is decreasing in a and in b; over a box where $a \in [a_-, a_+]$ and $b \in [b_-, b_+]$ every root lies in $[x^*(a_+, b_+), x^*(a_-, b_-)]$. A floating-point Newton iteration only locates each end; the end counts when h is proved negative just below it (a lower bound) or positive just above it (an upper bound), in outward-rounded interval arithmetic with the logarithm from a series with its remainder. Every other quantity is an interval extension; the rule $P_{wh} \leq 360$ bar is decided over the envelope by bisection, and a refusal requires two proved corners, one on each side. Decimal inputs are read as the exact rationals they denote, widened by one unit in the last place each way where the double is not exact.

3.3. Thresholds

When the rule is refused, the procedure bisects on a declared quantity with the envelope otherwise unchanged and reports the largest value at which the rule is PROVED over the whole box and the smallest at which it is REFUTED; here on the pump discharge P_d and on the lower end of the

roughness interval, the latter being a quantity an inspection could narrow. The thresholds are the product of a refusal: they turn an open question into a measurement or an operating change with a known effect on the verdict.

3.4. Controls

A battery of 24 checks runs before any receipt is published, with 9 red controls that must fire: a claim a tenth of a bar outside the enclosure, a manifold off by one cubic metre per day, a friction factor just outside the proved solutions, a velocity just outside Q/A, a flow just below the turbulent regime, a limit lowered under a proved violating corner, and a "gate" that evaluates the float model at the centre of the box — which is what a point pipeline does — each of which the battery must refuse. A second implementation in Python's decimal arithmetic at fifty digits, Colebrook solved by bisection on its own bracket, every constant written out again and no code shared, computes P_{wh} at the 64 corners and 24 random points of two operating points (176 inputs); the gate's thin-box interval contains the reference at every one, the tightest margin being 9.0×10^{-12} bar, and so does the whole-box enclosure. Every published receipt is re-decided in the reader's browser by the same code that wrote the record, and compared with it; the scenarios and the modules are pinned by SHA-256 (199bb6f8170f, e64151a67ae8).

4. Demonstration

4.1. The envelope and the proposals

Table 1 gives the declared envelope — typical, illustrative values, not any real line — and the two rules: a wellhead pressure of at most 360 bar and a velocity of at most 5 m/s. Nine proposals were decided (Table 2): two recommendations a model could make, and seven reference outputs each with one injected fault of a kind a verifier must catch — a catalogue diameter used in place of the surveyed one, a manifold that creates water, a solver that stopped before converging, a pressure the physics forbids, a unit conversion, an asserted velocity the inputs cannot give, and a correlation used outside its regime. Every receipt is what its scenario says it should be; no fault passes: 6 of the 7 are REFUTED and 1 is REFUSED (the laminar fall-off test, where the declared model does not apply and the procedure says so rather than decide).

Table 1: The declared envelope of the demonstrator's flowline (illustrative values) and its two rules.

input	declared interval	unit
length L	5,990–6,010	m
inner diameter D	152.0–152.8	mm
roughness ε	0.020–0.080	mm
water density ρ	1,022–1,028	kg/m^3
viscosity μ	1.08–1.39	mPa s
depth Δz	2,095–2,105	m
rule 1	$P_{wh} \leq 360$	bar
rule 2	$v \leq 5$	m/s

Table 2: The nine proposals decided over the envelope of Table 1: the operating point, the claimed wellhead pressure, the proved enclosure of P_{wh} over the envelope, the verdict of the receipt, and the number of checks proved.

proposal	Q (m^3/d)	P_d (bar)	claimed P_{wh}	enclosure (bar)	verdict	proved
proposal, +4.2% injection	7,000	206	352	[343.7, 362.0]	RECUSADO	4/5
the same flow, P_d lowered	7,000	200	346	[337.7, 356.0]	PROVADO	5/5
wrong parameter (catalogue diameter)	6,000	180	350.9	[336.6, 350.2]	REFUTADO	4/5
mass balance violated at the manifold	6,000	180	344	[336.6, 350.2]	REFUTADO	5/6
solver stopped early (friction factor)	6,000	180	358.9	[336.6, 350.2]	REFUTADO	4/6
boundary condition the physics forbids	6,000	180	395	[336.6, 350.2]	REFUTADO	3/5
unit conversion (psi labelled bar)	6,000	180	4989	[336.6, 350.2]	REFUTADO	3/5
a velocity the inputs cannot give	6,000	180	344	[336.6, 350.2]	REFUTADO	5/6
correlation outside its regime (laminar flow)	40	180	391	–	RECUSADO	2/3

4.2. The recommendation

A model recommends raising injection from 6,720 to 7,000 m^3/d (+4.2%), raising the pump discharge to 206 bar, and predicts 352 bar at the wellhead, below the limit, with a stated confidence of 97%. Over the declared envelope the wellhead pressure lies in [343.7, 362.0] bar: the claim is consistent with the model, the physical bound, the domain and the velocity rule are PROVED, and the pressure rule is REFUSED. The envelope holds an admissible input — the smoothest pipe, the least viscous and densest water, the largest diameter, the deepest water — at which $P_{wh} \geq 361.6$ bar, and another at which $P_{wh} \leq 344.2$ bar, each proved (Table 3). The receipt then publishes what decides: with $P_d \leq 204.0$ bar the rule is PROVED over the whole envelope, above 222.3 bar it is REFUTED; or an inspection showing a roughness of at least 27.000 mm (the envelope declares 0.020 to 0.080 mm) makes it PROVED at the proposed discharge. The second proposal, the same flow at 200 bar, is PROVED: $P_{wh} \in [337.7, 356.0]$ bar for every declared input (Fig. 2). The engineer who must act has, in place of "352 bar, 97%", a verdict, two inputs to check and a set point that would decide.

Table 3: The two witnesses of the refusal: an admissible input at which the proposed operating point violates the rule, and one at which it respects it, each with its wellhead pressure proved.

input	violates ($P_{wh} \geq 361.6$ bar)	respects ($P_{wh} \leq 344.2$ bar)
roughness ε (mm)	0.020	0.080
viscosity μ (mPa s)	1.08	1.39
inner diameter D (mm)	152.8	152.0
density ρ (kg/m^3)	1,028	1,022
depth Δz (m)	2,105	2,095
length L (m)	5,990	6,010

4.3. A corner recomputed by hand

The violating witness can be recomputed line by line in a spreadsheet: $Q = 7{,}000$ m^3/d and $P_d = 206$ bar at $\varepsilon = 0.020$ mm, $\mu = 1.08$ mPa s, $D = 152.8$ mm, $\rho = 1\,028$ kg/m^3, $\Delta z = 2\,105$ m and $L = 5\,990$ m give $A = 0.01834$ m^2 and $v = 4.418$ m/s, Re = 642,600, $x = 8.336$ and $f = 0.01439$ from Colebrook, a friction loss of 56.6 bar and a hydrostatic gain of 212.2 bar, hence $P_{wh} = 361.6$ bar, which the gate's enclosure at that point contains. The promise that a receipt can be checked without the authors' code is kept at the level of a hand calculation.

4.4. Monte Carlo face to face

The study an uncertainty-quantification workflow would run on the same recommendation draws inputs from the envelope and counts exceedances. With 10,000 uniform draws, 30 exceed the limit (0.3%; with 1,000 draws, 0.2%), the largest draw reaching 360.7 bar and the smallest 345.1 bar. The draws do find the exceedance; the difference lies elsewhere. First, the worst draw underestimates the worst case: the proved corner is 361.6 bar, 0.9 bar above the largest of 10,000 draws, because the violating region is a corner of a six-dimensional box that uniform sampling rarely visits. Second, the fraction 0.3% is a statement about a uniform prior over the envelope that nobody declared; the procedure's verdict, its witnesses and its thresholds do not depend on any prior. A study that reported "0.3% risk" would be reporting a property of its sampler; a study that reported "REFUSED, with a proved violating input and a set point that decides" would be reporting a property of the declared evidence. The two are complementary — the fraction of the envelope that violates is useful once the envelope is a distribution — and the receipt records the Monte Carlo result beside the verdict, labelled as statistical.

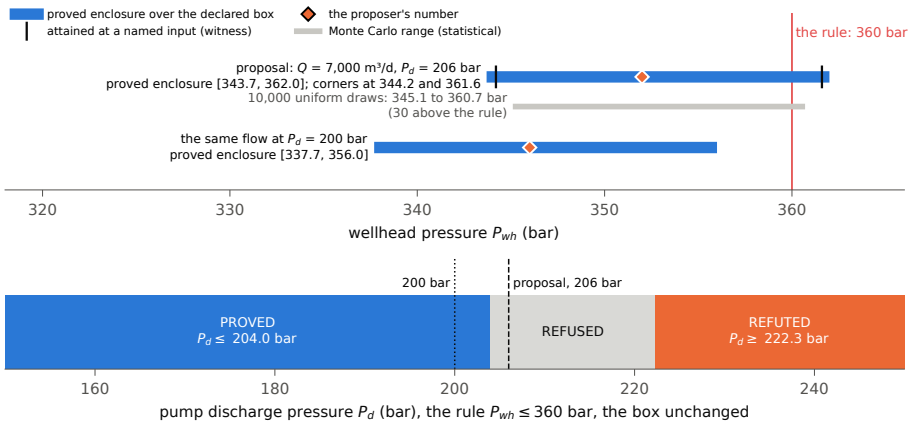


Figure 2: Top: the recommendation and its correction on the wellhead-pressure axis — the proved enclosure over the declared envelope, the two witnesses, the proposer's number, and the range of 10,000 uniform draws of the envelope, against the rule. Bottom: the verdict of the pressure rule as a function of the pump discharge with the envelope unchanged, and the thresholds the receipt publishes.

5. The same grammar on numbers nobody produced for a demonstrator

A decision procedure for proposed numbers is only as useful as its reach beyond a demonstrator. The companion paper (Toledo, 2026) applies the same grammar — proved, refused, decided, with witnesses and a public record — to the chain that produces a design wave: the extreme-value method of Reis et al. (2026), six families fitted by maximum likelihood to significant wave height at four block sizes, ranked by four criteria, with 100- and 1000-year levels, repeated on the method's own public hindcast. Three results of that work are the evidence that the grammar decides numbers of consequence. Over 2,589 open-sea cells, 46,602 fits were attempted and 40,778 are proved to be the likelihood's one maximum in a recorded box; 5,751 are refused with the proof that the family's likelihood is highest at a limit the optimiser can only approach, where a float routine prints a point on the way as the fit. The same scientific library (SciPy 1.18.1; Virtanen et al., 2020) called two ways on the same 175,320 hourly records prints 7.08 m or 15.20 m for the 100-year wave; the decision names which is the maximum it claims to be — the second — and measures the first as 19,850 log-likelihood units below a member of its own family; across 60 default calls none agrees with the certified maximum, and with the location parameter

fixed 45 do. And a published design table for five South Atlantic wind lease areas (Bhaskaran et al., 2023), printed from a commercial hindcast that cannot be re-run outside it, is decided row by row against the nearest cell of the public record (31 to 61 km away): 19 of 20 rows are off the maximum, by between −2.03 and +3.14 m in the 100-year wave — the distance between two records, made a number per row.

Before the engineering domains the procedure was calibrated where a proposed number is hardest to check by eye: published mathematics produced with or by frontier language models, decided in public with every verifier available. The register holds 131 decided claims, among them a benchmark certificate that would have improved the diagonal Ramsey bound, refuted because the benchmark's own checker accepted a construction when either of two required orientations passed (HorizonMath, 2026); a "preliminary, unverified" iteration printed in a published paper, proved by two independent programs in two languages agreeing to 25 digits (Gupta et al., 2024); and a library of 14 machine-found counterexamples of which 9 are certified whole and 5 in part (Sra, 2026). The strongest evidence that records of this kind re-certify is external: on 30 September 2026 an outside group re-derived every computational step of the project's proof of the value of $\lambda(4)$ in Chowla's cosine problem with their own code, re-certified all 2 231 finite cases and swept every reduced four-element set with maximum element at most 80 without finding a refuter (rainrzk, 2026). Nothing in that audit used the project's code; that is what a re-runnable record is for.

6. Relation to existing practice

Table 4 places the procedure. Validated numerics is the method it applies, not its contribution: interval Newton and Krawczyk operators, monotonicity tests and branch-and-bound are textbook (Moore et al., 2009; Hansen and Walster, 2004; Tucker, 2011), and probability-bounds analysis already propagates intervals through engineering models (Ferson and Ginzburg, 1996). The procedure differs in what it returns and in what it refuses to return. It returns a verdict about an output under declared conditions, with the two inputs that justify a refusal and the threshold that would remove it; it refuses to return a probability it has not been given a distribution for; and it records enough that a second implementation sharing no code can repeat it. Verification and validation standards (ASME, 2009; NASA, 2016) describe how to establish a model's credibility and how to report its uncertainty; the receipt is the artefact such a standard would want for each number the model then proposes. In process-safety regimes

Table 4: What existing instruments deliver about a model and what the decision procedure adds about one output under the declared conditions. Nothing is replaced; the procedure is a gate beside the model and an annex to its report.

instrument	delivers	the procedure adds
code and model verification and validation (ASME, 2009; NASA, 2016; Oberkampf and Roy, 2010)	credibility of the model in a domain; numerical and validation uncertainty	a decision about this output, under these declared inputs: the proved enclosure and the verdict
uncertainty quantification, confidence intervals, probability boxes (Ferson and Ginzburg, 1996; Beer et al., 2013)	a dispersion, or bounds on a distribution, under a declared input description	whether the output respects its equations and the rules over the whole envelope; the witnesses the sampler did not draw; no prior
a second calculation in another code	a second number	the arbiter: which number is the maximum (or the solution) it claims to be, and which stopped short
peer or classification-society review	an expert judgement	the mechanical re-derivation of each number as an input to the review, not a substitute for it
management of change (ANP, 2007)	the record of the change	the receipt as the technical record: model, envelope, verdict, thresholds, re-runnable at audit
formal verification (proof assistants)	a complete proof of a program	out of reach for today's engineering pipelines; the procedure proves the number, not the program

a change to an operating specification already requires a technical record (ANP, 2007); the receipt is one that can be re-run at audit, months later, from a single file.

7. Discussion

What a verdict means. A PROVED speaks for the declared model and the declared envelope: it says that no input the engineer declared makes the declared physics violate the declared rule. It does not say that the physics is right, that the envelope is wide enough, or that a regime the model excludes is absent. The pilot was chosen so that the last is a design choice — single-phase, steady, isothermal water injection is covered whole — and extensions to transients, multiphase flow, thermal profiles or erosion as a model enter as further declared models with their own envelopes and their own rule-or-flag classification, never as silent widenings of this one.

What a refusal buys. A refusal is the procedure's sharpest output. It measures the opacity of the declared evidence and prices what would remove it: a set point, an inspection, a measurement to a stated resolution. In the demonstrator the recommendation is REFUSED at 206 bar and PROVED at 200 bar; a line inspection narrowing the roughness to at least 27.000 mm would prove it as proposed. The thresholds are an incentive for disclosure: the proposer who wants a PROVED knows exactly what to declare.

What is at stake. The cost chain of a wrong operating set point is not the production of a floating unit; it is overpressure, an integrity question on the line, a shut-in of the injector and deferred pressure support, and the managed change that the regime requires in any case. The larger and more defensible number is the design basis: a 100-year wave fixes the design of hulls, moorings and risers (ISO, 2015; DNV, 2021), and the companion paper shows that the chain producing it returns, for the same data and the same library, two numbers a factor of two apart with no warning. The economic value of a verdict in either chain is not estimated here; it is a question for the operators who would use it.

Cost. A receipt is decided in milliseconds in a browser; the thresholds by bisection in under a second; the battery and the reference implementation in a few seconds. The arithmetic is IEEE-754 doubles with simulated directed rounding, one unit in the last place outward per operation.

8. Conclusions

A number a model proposes can be decided, not estimated, over the envelope the engineer declares: PROVED for every declared input, REFUTED with a proof, or REFUSED with an input on each side and the threshold that would decide. On a water-injection flowline the procedure refuses a plausible recommendation that a sampling study would have reported as a 0.3% risk, names the admissible input that violates the limit by a margin the sampler never reached, and names the set point that proves it; it refutes or refuses seven injected faults with the reason; and its receipts are re-run by a second implementation and by any reader. The same grammar decides maximum-likelihood fits over a public hindcast, a library's two answers for one design wave, and a published design table. What is proposed to the reliability community is a small vocabulary and a discipline: a verdict with a witness, refusal as a result, and a record that travels without the code that made it.

Declaration of competing interest

The author declares no competing financial or personal interests. The demonstrator uses illustrative values and no operator's data.

Data availability

The gate, its scenarios, its battery and its reference implementation are public in the cert-machine repository (`https://github.com/carlostoledo1891/cert-machine`, commit aae61990d737) under `apps/contraprova` (MIT); the receipts and the Monte Carlo record are `apps/contraprova/data/gate-ledger.json`; the live demonstrator, in which every receipt is re-decided in the browser, is at `https://carlostoledo.co/contraprova/`. The extreme-value records are those of the companion paper; the register of decided mathematical claims is `certs/claims-ledger.json`.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used large language models (Claude, Anthropic) to draft code, exposition and this manuscript in a certified-verification setting: constructions and prose were proposed with the help of the models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for a record to be accepted. After using the tool the author reviewed and edited the content and takes full responsibility for the content of the publication. The records, not the models, are the evidence offered; they are public and re-runnable.

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References

Agência Nacional do Petróleo, Gás Natural e Biocombustíveis, 2007. Resolução ANP n.° 43, de 6 de dezembro de 2007: Regulamento Técnico do Sistema de Gerenciamento da Segurança Operacional (SGSO), práticas de gestão 13 (integridade mecânica) e 16 (gerenciamento de mudanças).

American Petroleum Institute, 1991. API RP 14E: Recommended practice for design and installation of offshore production platform piping systems, 5th ed.

ASME, 2009. V&V 20-2009 (R2016): Standard for verification and validation in computational fluid dynamics and heat transfer. American Society of Mechanical Engineers, New York.

Beer, M., Ferson, S., Kreinovich, V., 2013. Imprecise probabilities in engineering analyses. Mechanical Systems and Signal Processing 37, 4–29.

Bhaskaran, S.K., Verma, A.S., Goupee, A.J., Bhattacharya, S., Nejad, A.R., Shi, W., 2023. Comparison of extreme wind and waves using different statistical methods in 40 offshore wind energy lease areas worldwide. Energies 16, 6935.

Colebrook, C.F., 1939. Turbulent flow in pipes, with particular reference to the transition region between the smooth and rough pipe laws. Journal of the Institution of Civil Engineers 11, 133–156.

DNV, 2021. DNV-RP-C205: Environmental conditions and environmental loads. Recommended practice.

Ferson, S., Ginzburg, L.R., 1996. Different methods are needed to propagate ignorance and variability. Reliability Engineering & System Safety 54, 133–144.

Gupta, P., Ndiaye, N., Norin, S., Wei, L., 2024. Optimizing the CGMS upper bound on Ramsey numbers. arXiv:2407.19026.

Hansen, E., Walster, G.W., 2004. Global Optimization Using Interval Analysis, 2nd ed. Marcel Dekker, New York.

HorizonMath, 2026. The HorizonMath benchmark, arXiv:2603.15617v2 (10 September 2026), Appendix A; code at github.com/ewang26/HorizonMath, commit 4ef0b61a.

ISO, 2015. ISO 19901-1:2015 Petroleum and natural gas industries — Specific requirements for offshore structures — Part 1: Metocean design and operating considerations.

Jaulin, L., Kieffer, M., Didrit, O., Walter, É., 2001. Applied Interval Analysis. Springer, London.

Moore, R.E., Kearfott, R.B., Cloud, M.J., 2009. Introduction to Interval Analysis. SIAM, Philadelphia.

NASA, 2016. NASA-STD-7009A: Standard for models and simulations. National Aeronautics and Space Administration.

Oberkampf, W.L., Roy, C.J., 2010. Verification and Validation in Scientific Computing. Cambridge University Press, Cambridge.

Petrobras, 2026. Digital twins na Petrobras impulsionam eficiência operacional. Nossa Energia (corporate publication).

Agência Petrobras, 2026. Petrobras usará gêmeo digital para otimizar produção e escoamento de petróleo (Jubarte pilot).

rainrzk, 2026. An independent audit of the $\lambda(4)$ proof. Repository erdos510-lambda4-audit; report on teorth/erdosproblems, issue 392, 30 September 2026.

Reis, R.A.N., Guimarães, P.V., Farina, L., Paul, S., de Paula, T.P., Ribeiro, E.O., 2026. Global assessment of extreme value analysis for significant wave height. Ocean Engineering 359, 125841.

Roy, C.J., Oberkampf, W.L., 2011. A comprehensive framework for verification, validation, and uncertainty quantification in scientific computing. Computer Methods in Applied Mechanics and Engineering 200, 2131–2144.

Rump, S.M., 2010. Verification methods: rigorous results using floating-point arithmetic. Acta Numerica 19, 287–449.

Sra, S., 2026. GPT, the counterexample machine. arXiv:2608.29595; the library at github.com/suvrit/count-ex-machina, commit dbf67374.

Toledo, C., 2026. Certified return levels: deciding maximum-likelihood extreme-value fits of significant wave height over a public global hindcast. Companion manuscript, in preparation.

Tucker, W., 2011. Validated Numerics: A Short Introduction to Rigorous Computations. Princeton University Press, Princeton.

Virtanen, P., Gommers, R., Oliphant, T.E., et al., 2020. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nature Methods 17, 261–272.